AJITESH VARDHAN

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Profile

A driven and technically capable Mechanical Engineering undergraduate at Loughborough University, seeking a 12-month industrial placement in product design and development. Experienced in the full design cycle from concept and CAD modelling to prototyping, physical testing, and iteration, including ongoing industry-based work with Rolls-Royce Submarines. Proficient in Siemens NX (3D modelling, FEA, NX CAM) and MATLAB, with additional skills in Python and Excel-based numerical modelling. Committed to delivering robust, well-tested engineering solutions in collaborative team environments.

Education

Loughborough University | Loughborough, England **2024 – 2028**
BEng Mechanical Engineering with Placement Year
Grade: 2:2 (First Year)

Reading School | Reading, England **2017 – 2024**
A-Levels: Mathematics (B), Physics (B), Computer Science (B)
GCSEs: 10 GCSEs including Mathematics (8), English Language (6) and Sciences (8, 7, 7)

Projects

Desktop Wind Tunnel | Personal Project **April 2026 - Ongoing**

- Designed a wind tunnel with a 4:1 Area Contraction Ratio and an integrated flow straightener, utilizing a Bell-mouth inlet geometry to accelerate airflow while maintaining a uniform velocity profile
- Overcame 180mm³ build volume constraints by engineering a multi-part interlocking assembly, utilizing tongue-and-groove joints to maintain structural alignment and flow continuity
- Developed a custom aerodynamic balance utilizing a strain-gauge load cell and a 24-bit HX711 ADC to quantify drag forces on 1/64 model cars
- Designed a custom dual-layer Printed Circuit Board (PCB) using EasyEDA to automate wind tunnel data acquisition and fan speed regulation.
- Performed through-hole soldering and component population, troubleshooting hardware-software interfaces to ensure system reliability under testing conditions.

CADMAT Competition | Loughborough University **February 2026 – Ongoing**

- Designed and tested a load-bearing acrylic bracket, optimising for maximum load capacity (up to 30kg) while minimising mass and deflection.
- Utilised Siemens NX for 3D solid modelling and Finite Element Analysis (FEA) to validate and iterate virtual prototypes, identifying stress concentrations and failure points prior to manufacture.
- Performed hand calculations to independently verify FEA results, ensuring confidence in simulation outputs and reinforcing analytical understanding of structural behaviour.
- Generated CNC toolpaths using NX CAM to manufacture prototypes, performing physical tests and manual modifications to refine performance across multiple competition rounds.
- Compiled a formal report and delivered a presentation documenting design rationale, FEA findings, and test outcomes, while collaborating in a team of 4 to manage the full design cycle under time pressure.

IBP – Venturi| Rolls-Royce + Loughborough University **October 2025 – March 2026**

- Designed a test rig to measure Venturi differential pressure and quantify the impact of throat diameter on flow rate measurement uncertainty.
- Developed a full CAD model of the Venturi assembly in Siemens NX, ensuring geometric accuracy across components to support reliable prototype manufacture and validation.
- Investigated flow rate uncertainty across various operating conditions using hand calculations and analytical modelling, recommending a throat diameter tolerance to minimise measurement error.
- Produced 3D-printed prototypes using FDM equipment to validate geometry and support physical testing, iterating on designs based on experimental outcomes.
- Liaised directly with Rolls-Royce engineers to align project scope with industry requirements and authored a formal technical report communicating findings to both academic and industry stakeholders.

Smart Campus Project – Detailed Design | Loughborough University **April 2025**

- Produced the lightest bridge in the cohort at 798.7g, exceeding structural requirements by withstanding 130kg – demonstrating effective weight-optimised design.
- Applied truss analysis to optimise the design for strength, weight, and cost, achieving the best structural efficiency score in the cohort.
- Developed detailed CAD models to define geometry, inform material usage, and guide the fabrication process.
- Conducted a cost analysis alongside structural performance metrics to balance competing design constraints within a fixed budget.
- Collaborated within a team to divide responsibilities across design, analysis, and build phases, coordinating effectively to meet project deadlines.

Sprint-Based Trebuchet Project | Loughborough University **October 2024**

- Acted as Project Lead for the full design, build, and test cycle within a one-week sprint, prioritising tasks under time pressure to deliver a functional prototype on deadline.
- Used equations of motion in Excel to predict projectile trajectory and calculate the launch angles required to hit multiple targets accurately.
- Produced sketches and technical drawings to communicate design intent and coordinate the build across the team.
- Iterated the design following initial test launches, adjusting counterweight mass and sling length to improve range accuracy and consistency.
- Delegated tasks across team members based on individual strengths, maintaining momentum and ensuring all sub-assemblies were integrated on schedule.

Skills

CAD & Software

Siemens NX (3D modelling, FEA, NX CAM), MATLAB (ranked 8th in Exam), Microsoft Excel (numerical/trajectory modelling), Python (SQLite, API integration), GD&T (Geometric Dimensioning and Tolerancing), DFMA (Design for Manufacture and Assembly), EasyEDA

Extracurriculars

- LAMDA Grade 8 in Speaking with Distinction, enhancing skills in public speaking and presentation.
- Computer Science Mentor, tutoring GCSE students in programming fundamentals.
- Assistant coach for under-9s at Shinfield Cricket Club, developing group management and communication skills in a coaching environment.
- Scout Leader, led diverse groups during Eurocamp, developing team motivation, conflict resolution, and cross-cultural communication skills.